Note: Q 10 carry 6 marks while rest carry 1 each.

Select the best choice.

1. **Which of the following best describes mutually exclusive alternatives?**
   A. Alternatives that may be selected together
   B. Alternatives evaluated only using payback period
   C. Alternatives where choosing one eliminates the possibility of choosing another
   D. Alternatives with identical cash flows
2. **When evaluating mutually exclusive alternatives using PW or FW, the correct decision rule is:**
   A. Choose the alternative with the highest payback ratio
   B. Choose the alternative with the highest PW or FW
   C. Choose the alternative with the lowest initial cost only
   D. Choose all alternatives with positive NPWs
3. **For equal-life alternatives, which method is generally acceptable to compare economic worth?**
   A. Only payback period
   B. Only incremental ROR
   C. Either PW or FW, because they give the same ranking
   D. Only EUAW
4. **When comparing equal-life alternatives using PW, the alternative with the highest PW is:**
   A. Always the least risky
   B. Always the least-cost option
   C. The alternative that should be selected
   E. The alternative with the smallest salvage value
5. **When comparing alternatives with different useful lives, which adjustment is needed?**
   A.
   B. Normalize using simple payback
   C. Convert to a common study period
   D. Ignore salvage value
   E. Use only FW, not PW
6. **To compare different-life alternatives using PW correctly, one typically:**
   A. Uses only the higher-life alternative
   B. Repeats alternatives to their least common multiple lives
   C. Uses only the smaller life alternative
   D. Ignores replacement costs
7. **Capitalized cost in engineering economics refers to:**
   A. The present worth of all costs over a finite time period
   B. The future worth of all project expenses at the end of the study period
   C. The present worth of a project that will last indefinitely
   D. The annual cost of owning and operating an asset
8.
9. ***A project has an initial cost of* $500,000 and requires a perpetual annual maintenance cost of $20,000. If *the interest rate is 5%,* what is the capitalized cost of the project?**
   A.

B. \$500,000

C. \$900,000

D. \$500,000 + \$20,000 × 5 = \$600,000

E. D. \$500,000 + \$20,000 ÷ 0.05 = \$900,000

10. The Capitalized Cost of an asset with perpetual replacement cycles includes:

A. Only the first cost

B. First cost + PV of perpetual annual expenses

C. Only the annual operating cost

D. Only the salvage value of the first asset

11. Three months ago, a portion of a house was rented out for PKR 60,000 per month. At the tenant's request, an additional area has now been added, and both parties agreed to increase the rent by PKR 5,000, resulting in a new monthly rent of PKR 65,000. However, the tenant can only pay PKR 63,000 per month, creating a monthly shortfall of PKR 2,000. According to standard practice, the rent normally increases by 10% after the first year. The tenant is willing to accept a higher increase so that the total accumulated shortfall of year 1 is fully recovered through extra rent collected during Year 2. **Determine the revised annual rent increase percentage that will fully recover the Year-1 shortfall by the end of Year-2.**